

E4BCA1T1 – E – 26 – 7261



FOURTH SEMESTER B.C.A. (REVISED NEP) DEGREE EXAMINATION, JUNE 2026
COMPUTER GRAPHICS (DSC-1)

Time : 3 Hours]

[Max. Marks : 80

Instruction : Answer the following questions.

PART – A

Answer the following questions. **Each** question carries **2** marks. **(10×2=20)**

1. Define computer graphics.
2. List any four applications of computer graphics.
3. Define :
 - a) Aspect ratio
 - b) Frame buffer.
4. Define an output primitive.
5. Define scan conversion.
6. Define transformation.
7. What is clipping ? List types of clipping.
8. Define window.
9. List any four interactive display techniques.
10. List any four 3D display techniques.

PART – B

Answer **any six** questions. **Each** carries **5** marks. **(6×5=30)**

11. Explain the working of cathode ray tube with neat diagram.
12. Write a note on printer.

[P.T.O.]

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13. Write steps for Bresenham's line drawing algorithm.
14. Write a note on area fill attributes.
15. Explain and write matrix representation for basic 2D translation and scaling.
16. Derive window-to-viewport coordinate transformation equation.
17. Explain the following devices.
 - a) Keyboard
 - b) Joystick.
18. What is Bezier curve ? List the properties of Bezier curve.

PART – C

Answer **any three** questions. **Each** question carries **10** marks.

(3×10=30)

19. Explain raster scan and random scan display with diagram.
20.
 - a) Write the steps for mid-point circle generating algorithm.
 - b) Explain the following line attributes.
 - a) line type
 - b) line width.
21. Define polygon clipping. Explain Sutherland Hodgeman polygon clipping with diagram.
22. Explain parallel and perspective projection.